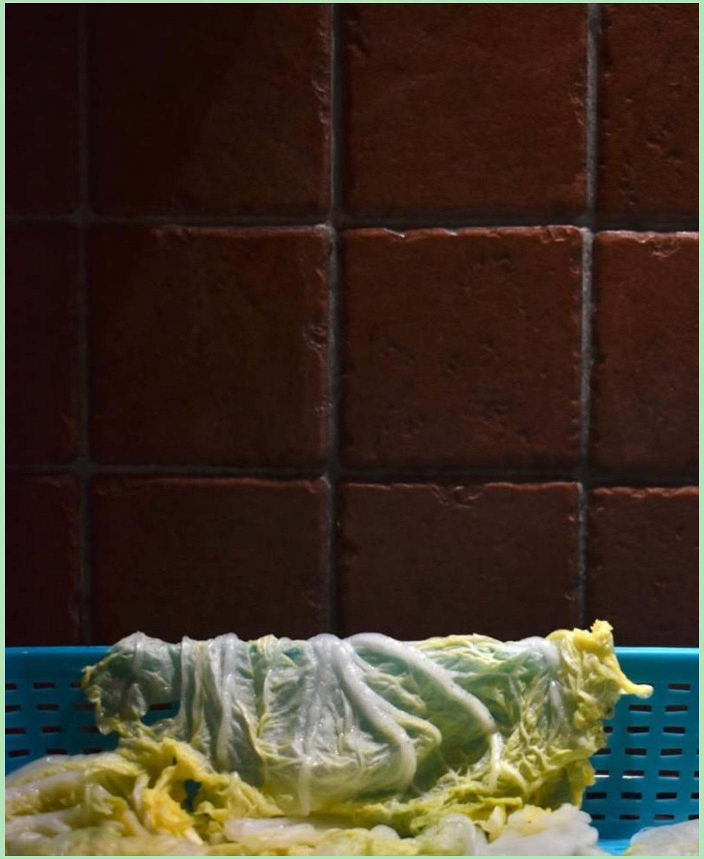

Veronica Yu

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- Harvard University Class of 2028
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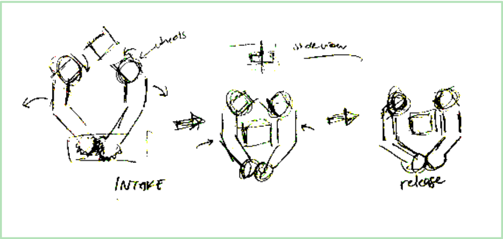
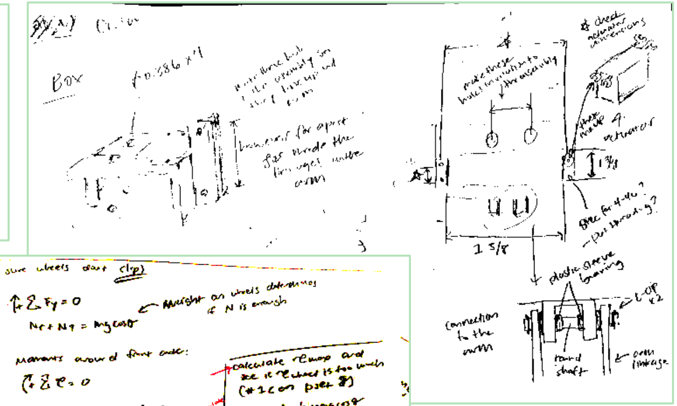
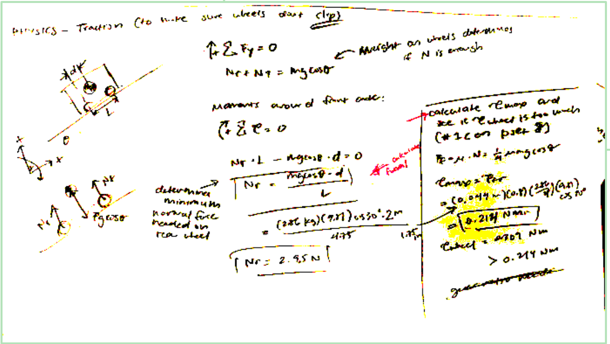
Competition field

Turf Wars Claw Robot

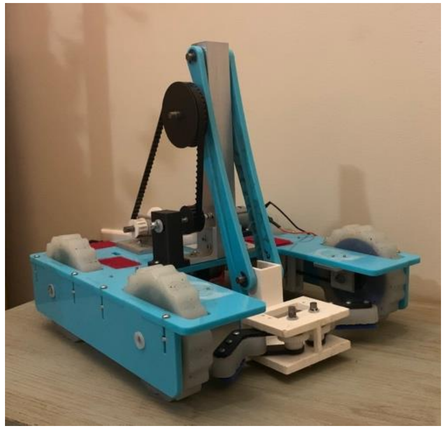
Class Project (Jan 2025 – May 2025)







Initial designs and prototypes



Final robot

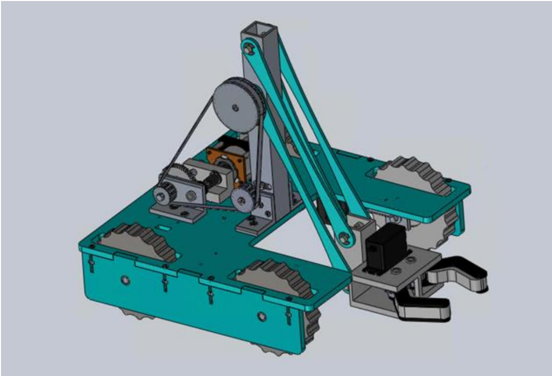
CONTEXT

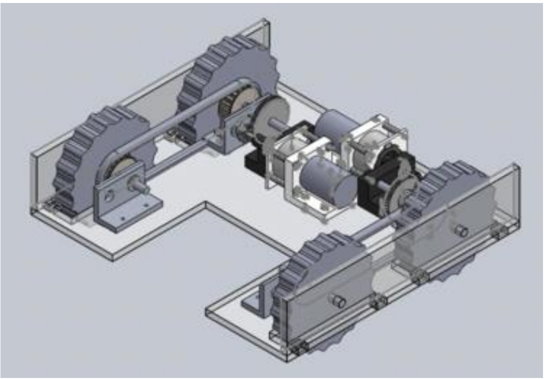
Robot challenges for competition:

- Pick up blocks, hockey pucks, and dog toys and place in tic-tac-toe board
- Traverse synthetic grass and 15 and 30 degree inclines

CONTRIBUTIONS

- Managed project tasks and reviewed all designs
- Designed drivetrain and completed 80% of robot CAD
- Calculated torque, motor, COM constraints
- Manufactured and assembled gearbox, arm, and claw components





SKILLS

- Mechanical Design
- SolidWorks
- CNC Milling
- Manual Lathing
- Bandsaw, Drill Press

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Wavelength

Conflux Collective – *Co-Project Lead*
(Sep 2025 – May 2026)

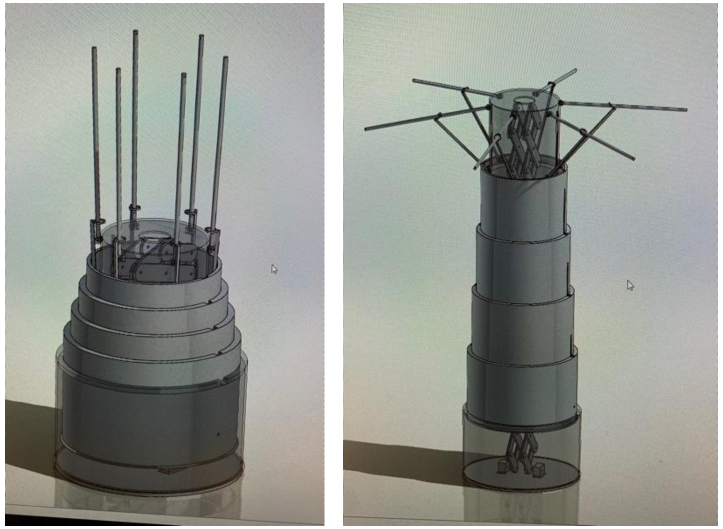
CONTEXT

What does it mean to be on the same wavelength?

Wavelength is an interactive art installation where two participants must hum together for the sculpture to blossom.

Components:

- A central telescoping tree
- A bed of kinetic flowers
- An ambient soundtrack that incorporates participants' voices

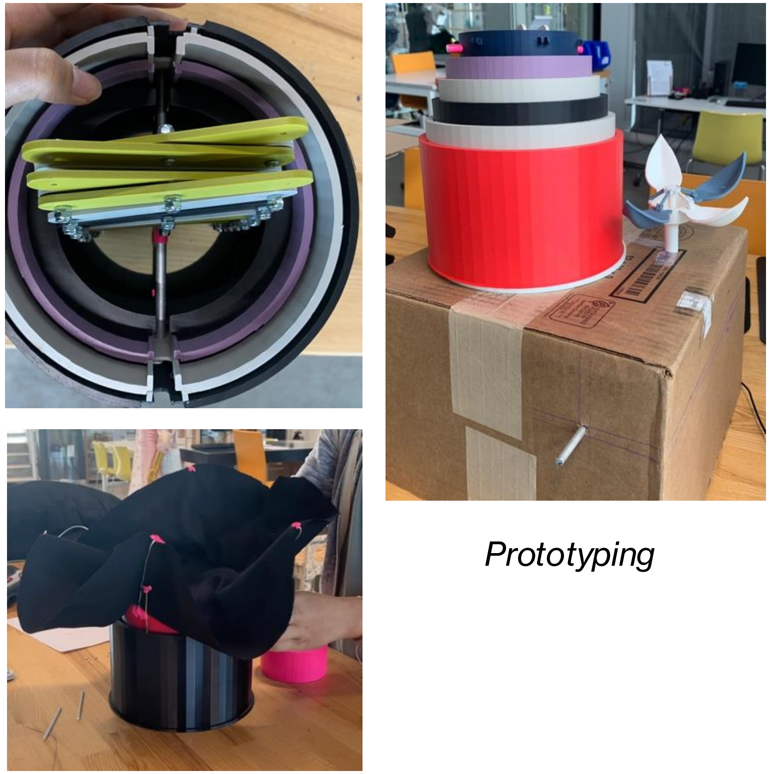


CAD of telescoping tree

CONTRIBUTIONS

- Prototyped telescoping cylinders powered by a scissor lift
- Integrated electronics to control motor speed with a Raspberry Pi based on singing from two microphones
- Designed testing to improve user-friendliness of interactive feedback

Testing of motor control and user interface through singing



Prototyping

SKILLS

- Prototyping
- Mechanical Design
- Raspberry Pi
- Electronics
- User Testing

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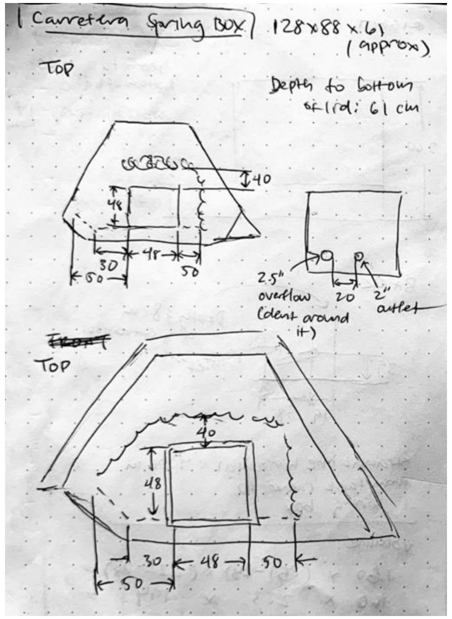
Gravity-Fed Water Systems

Engineers Without Borders – *Revit Technical Lead, 2025 & 2026 Travel Team Member* (Jan 2025 – present)



Dimensioning on the field

CONTEXT

9 year long project in Dominican Republic to build water distribution system (closed out in 2025)

- Gravity-fed water system with borehole well and two tanks

New project in Guatemala to create solar-powered water distribution system using spring sources

CONTRIBUTIONS

- Revise Revit models of tanks, valve boxes, and wellhouses
- Communicated with community about experience with water system
- Conducted water quality and water flow tests
- Compiled detailed notes of each travel day

Upper tank valve box

Upper tank

SKILLS

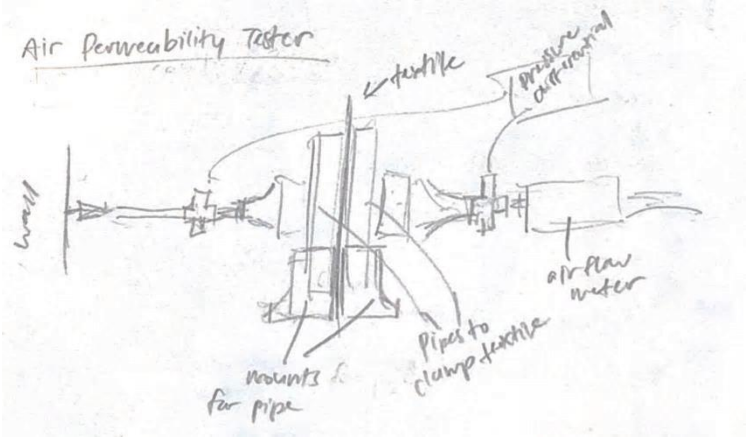
- Humanitarian Design
- International Development
- Revit
- Water Engineering

Air Permeability Tester

Research (Dec 2025 – Mar 2026)

CONTEXT

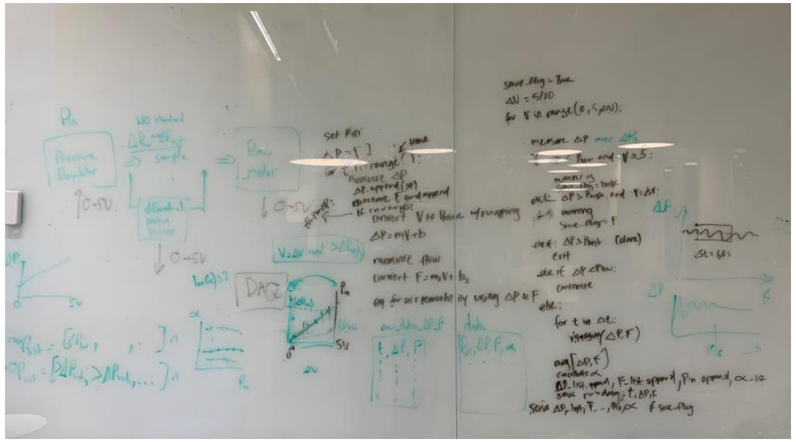
Build an air permeability tester that measures air flow rate through a material at a set pressure differential.



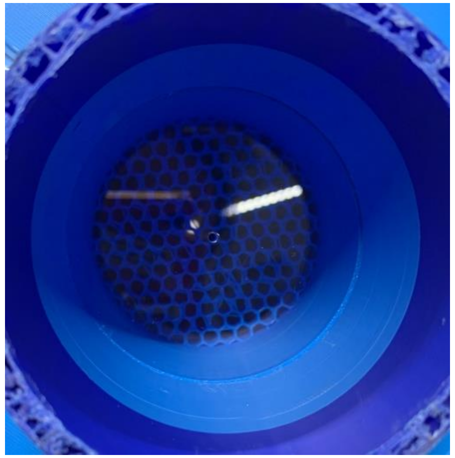
Concept drawing

CONTRIBUTIONS

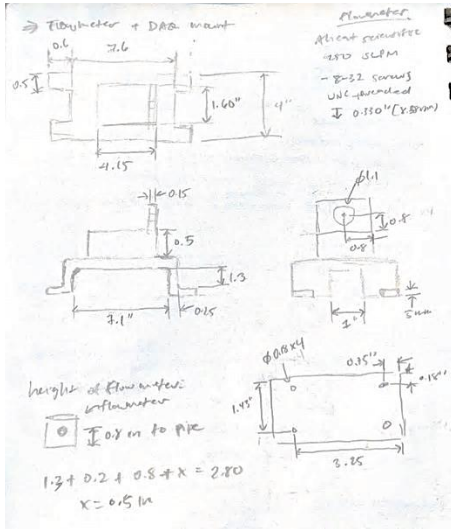
- Built mechanical fixtures to clamp specimen and connect tubing while preventing leakage
- Integrated electronics for pressure differential sensor and flowmeter
- Designed and executed calibration tests to align with theoretical calculations



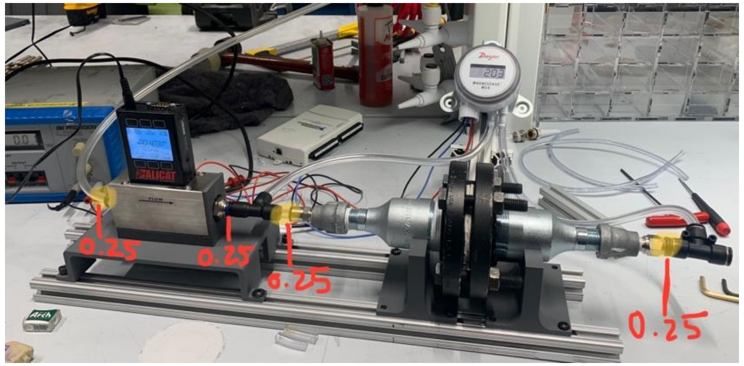
Initial planning for DAQ MATLAB integration

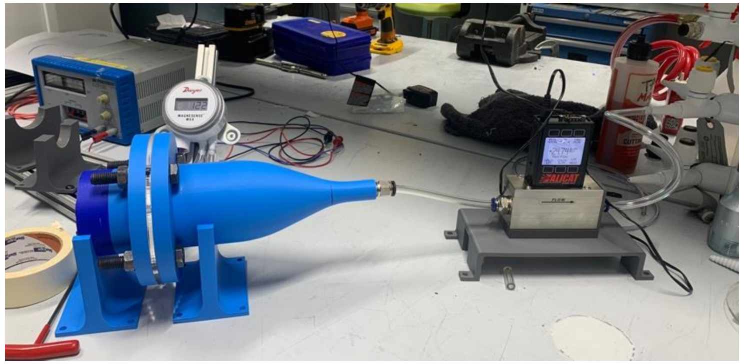


Inside view with orifice plate



Flowmeter and DAQ mount





Testing setup

SKILLS

- Sensor Integration
- Mechanical Design
- SolidWorks
- 3D Printing
- Testing Validation

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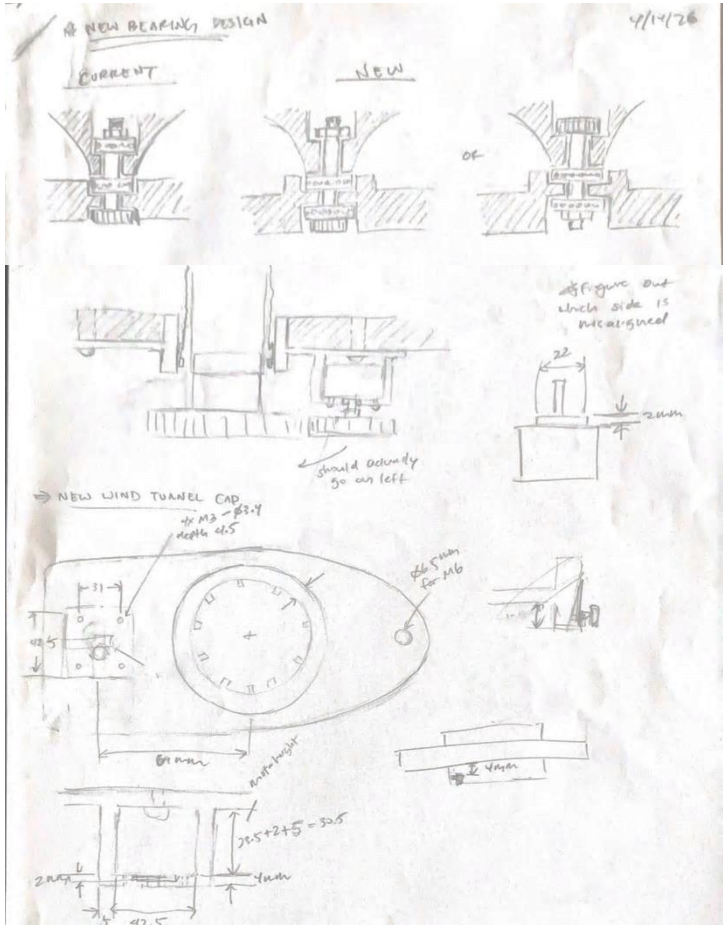
Rotational Fixture

Research (Mar – May 2026)

CONTEXT

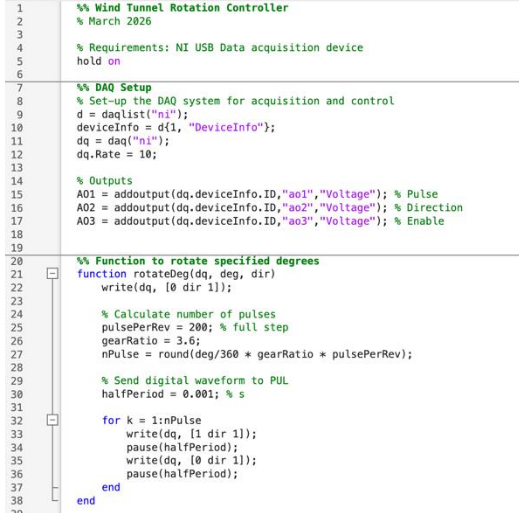
Build a fixture to rotate a cylinder in a wind tunnel test section from 0º to 180º in 5º increments.



Bearing and motor mount designs

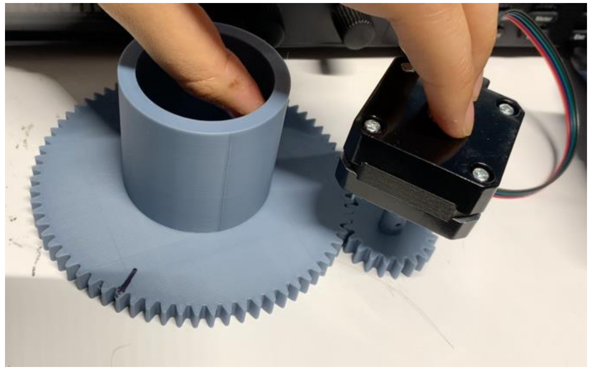
CONTRIBUTIONS

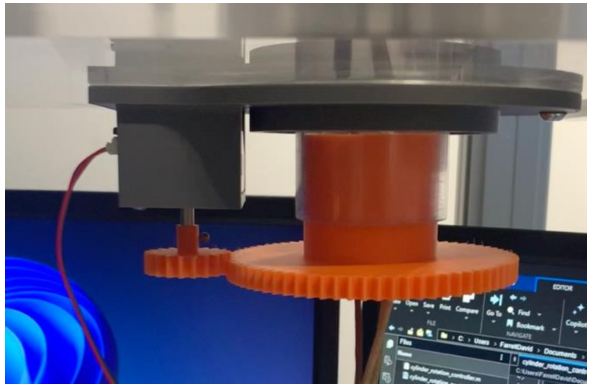
- Selected stepper motor and motor controller, reading through specifications
- Designed motor mount and bearing setup compatible with wind tunnel and cylinder
- Wrote MATLAB script to control rotation amount through wind tunnel DAQ



```
%% Wind Tunnel Rotation Controller
% March 2026

% Requirements: NI USB Data acquisition device
hold on

%% DAQ Setup
% Set-up the DAQ system for acquisition and control
d = daqlist("ni");
deviceInfo = d{1, "DeviceInfo"};
dq = daq("ni");
dq.Rate = 10;

% Outputs
A01 = addoutput(dq.deviceInfo.ID,"ao1","Voltage"); % Pulse
A02 = addoutput(dq.deviceInfo.ID,"ao2","Voltage"); % Direction
A03 = addoutput(dq.deviceInfo.ID,"ao3","Voltage"); % Enable


%% Function to rotate specified degrees
function rotateDeg(dq, deg, dir)
    write(dq, [0 dir 1]);

    % Calculate number of pulses
    pulsePerRev = 200; % full step
    gearRatio = 3.6;
    nPulse = round(deg/360 * gearRatio * pulsePerRev);

    % Send digital waveform to PUL
    halfPeriod = 0.001; % s

    for k = 1:nPulse
        write(dq, [1 dir 1]);
        pause(halfPeriod);
        write(dq, [0 dir 1]);
        pause(halfPeriod);
    end
end
```

MATLAB script





Rotational fixture

SKILLS

- Electronics Integration
- MATLAB
- Mechanical Design
- SolidWorks
- 3D Printing

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Python Image Tracking

Bertoldi Group Research (Jun – Aug 2025)

Harvard 2025 PRISE Fellow

CONTEXT

Research Question: Can kirigami-patterned textiles enhance convective cooling relative to a bare surface?

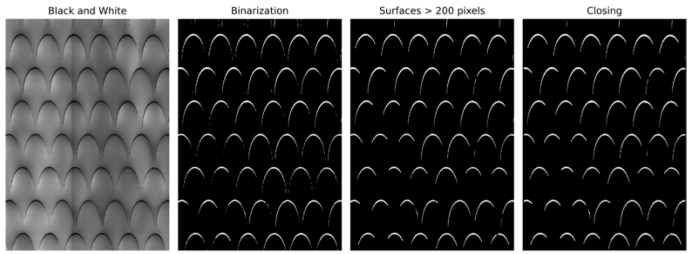
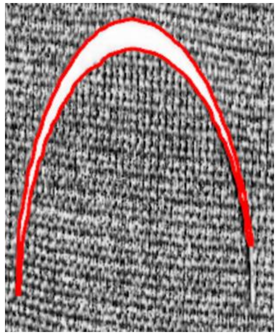
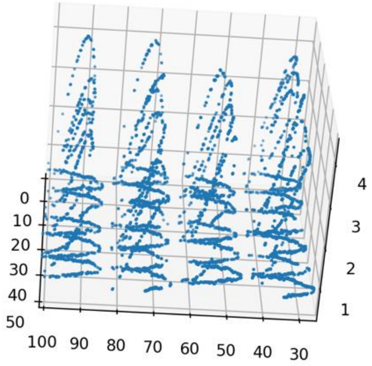


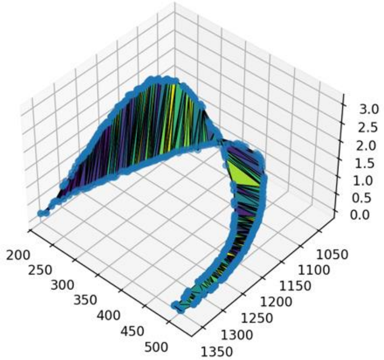
Image processing



Detecting contours



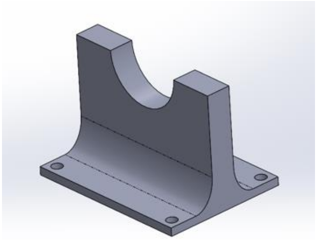
Adding height data

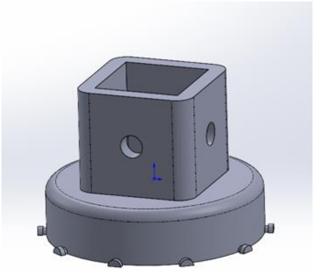


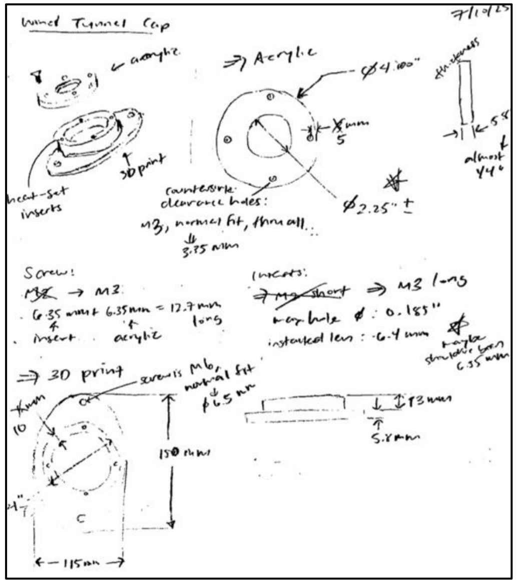
3D area

CONTRIBUTIONS

- Develop post-processing script to track contours as textile is stretched and calculate 2D area, 3D area, and volume
- Design parts for experimental setup
- Conduct Instron experiments to collected data with laser profilometer
- Fabricate textile samples







SKILLS

- Python
    - OpenCV, SciPy
- Mechanical Design
- SolidWorks
- Instron
- Laser Cutting

**Details and pictures of the textile are excluded due to ongoing research*

Table of Contents

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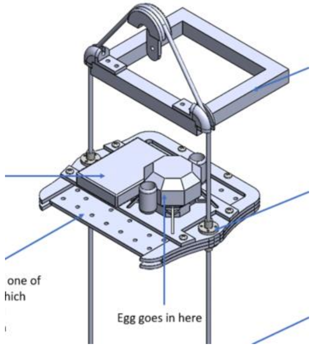
Egg Drop Contraption

Class Project (Nov - Dec 2025)

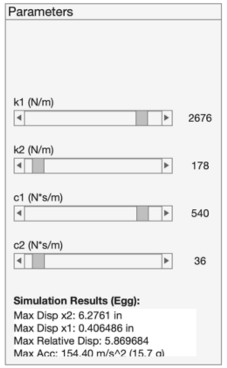
CONTEXT

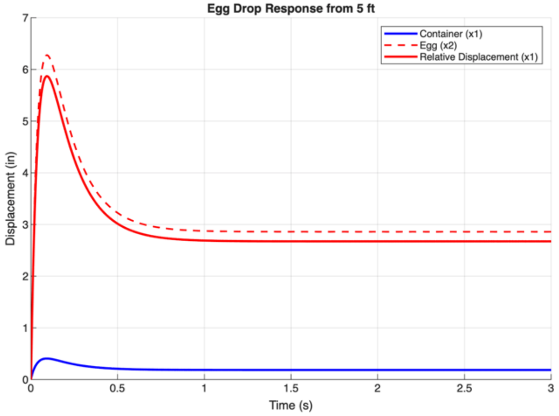
- Build contraption to connect to bottom of egg carriage and prevent egg cracking from the tallest height
- Use provided springs, dampers, and plywood



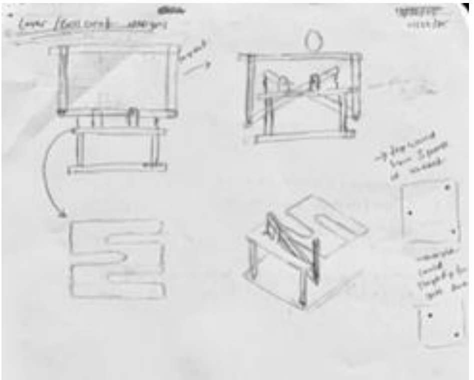
Provided shocks with max compression of 1.3 in

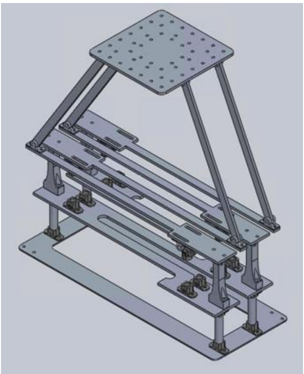
Egg carriage system

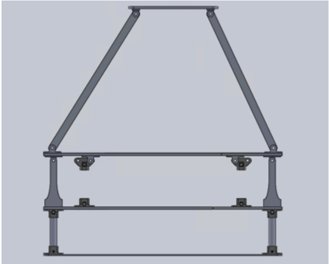


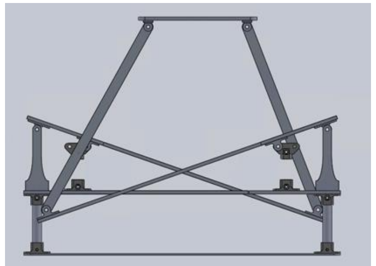


MATLAB Calculations









CONTRIBUTIONS

- Wrote MATLAB code to model response of 2-DOF system
- Designed lever system to lower effective spring and damping constants and prevent bottoming-out
- Created CAD and assembled system

Result:
3D prints immediately snapped – we should've flipped the printing orientation ☹

SKILLS

- Mechanical Design
- MATLAB
- SolidWorks
- 3D Printing
- Laser Cutting

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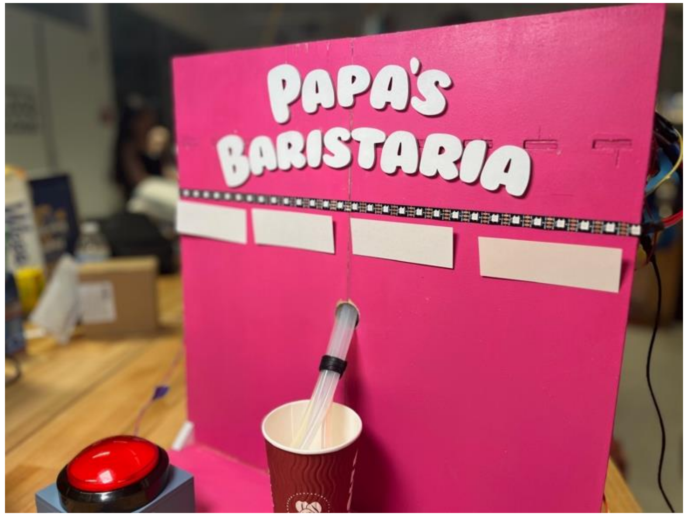
Papa’s Baristaria

Class Project (April – May 2026)



Final project front view

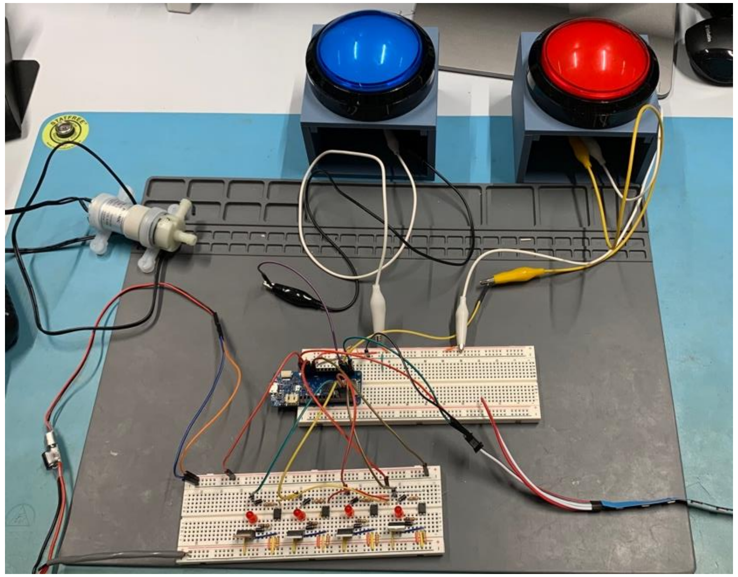
CONTEXT

Create an real-life version of Papa Louie games with two phases:

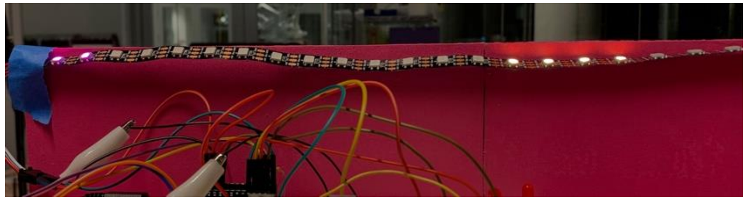
1. Two players compete to fill up their half of the LED strip first, with 3 button presses lighting up one LED
2. The winner stops the oscillating LED pattern to select a drink to be dispensed

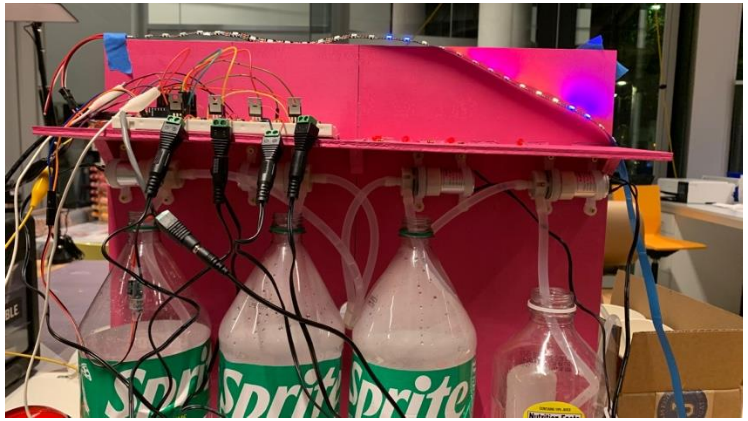
CONTRIBUTIONS

- Wrote code for Arduino control of LED strip and overall game logic
- Completed overall circuit connecting buttons, LED strip, and Arduino to circuit for pumps



Inside view with orifice plate





LED strip testing and back view

SKILLS

- Arduino
- Electronics Integration
- Circuit Debugging

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From Amazon to Atelier: Violin Workshopping

Class Project (Jan – May 2025)

CONTEXT

Improve the sound of a cheap, unfinished violin

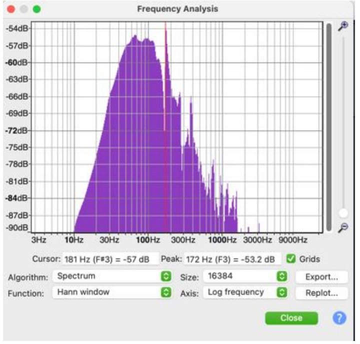
Beginning product

CONTRIBUTIONS

- Tested modes 2 and 5 using Audacity frequency analysis and tea
- Carved top plate to adjust modes
- Shaved bridge
- Adjusted soundpost to fit perfectly
- Assembled violin

Mode 2

Mode 5



End product

SKILLS

- Woodworking
    - Planes
    - Handsaw

Voted best sounding violin in class of 6 (through anonymous recordings)

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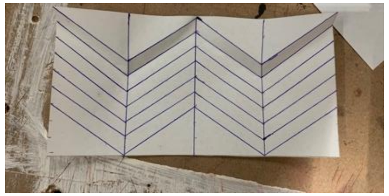


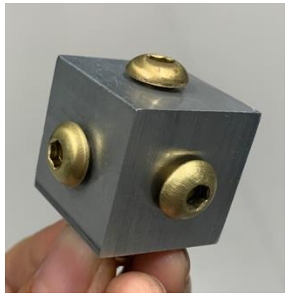
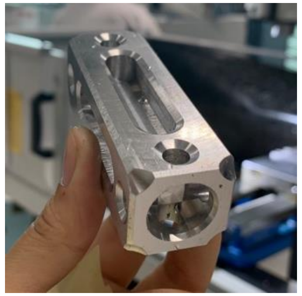

Art, Photography, Machining, Origami






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